



E-commerce and Digital Payments

What is E-Commerce?

E-Commerce (short for **electronic commerce**) is the **buying and selling of goods, services, or information** over the internet. It includes everything from online shopping, digital payments, and online banking, to auction sites and subscription services.

In simple words: **Shopping and business done online.** Instead of visiting a physical store, you open a website or app, pick what you want, pay digitally, and get it delivered to your door.

💡 **Why it matters:** *E-Commerce makes shopping faster, easier, and more convenient. It connects buyers and sellers from anywhere in the world, 24/7.*



Main Components of E-Commerce

These are the **building blocks** that make an online store work. Every e-commerce system relies on these parts to function smoothly.



Online Storefront

The **website or app** where customers browse products, read descriptions, and add items to their cart.

📱 Examples: Amazon, Shopify stores, eBay.



Payment Gateway

The system that **processes payments** securely. It connects the store to banks and verifies transactions.

🔒 Examples: PayPal, Stripe, Razorpay.



Inventory & Order Management

Tracks **stock levels, orders, and shipping**. Ensures that what you see online is actually available.



Customer Management

Handles **user accounts, profiles, order history**, and support. Makes shopping personal and repeatable.

 Automatically updates when a product is sold.

 Includes wish lists, reviews, and loyalty points.



Logistics & Delivery

The **physical movement** of goods from the warehouse to your doorstep. Includes tracking and returns.

 Often integrated with courier services.



Analytics & Marketing

Tools that **track visitor behavior, sales, and campaigns**. Helps businesses grow and improve.

 Email marketing, ads, and recommendations.

Elements of E-Commerce Security

Security is the **backbone** of online business. These five elements protect both the customer and the seller from fraud, data theft, and system failures.

01

Confidentiality

Ensures that **sensitive data** (passwords, credit cards, addresses)

02

Integrity

Guarantees that information is **accurate and unchanged** from the

is kept private and accessible only to authorized people.

🔑 Encryption (SSL/TLS) keeps data unreadable during transmission.

moment it's sent to when it's received.

✔️ Digital signatures and checksums verify that data hasn't been tampered.

03

Authentication

Confirms the **identity** of both the user and the website. Prevents impostors from accessing accounts.

🔑 Two-factor authentication (2FA), passwords, and biometrics.

04

Authorization

Determines **what a user is allowed to do** after they are authenticated. Not everyone gets full access.

👤 Admin vs. customer permissions; only admins can change prices.

Remember: These five pillars work together. If one fails, the whole system becomes vulnerable.

⚠️ E-Commerce Threats

Threats are dangers that can harm an online business or its customers. Here are the most common ones and how they work.

Phishing

Fraudsters send fake emails or messages pretending to be a trusted store. They trick you into giving away **passwords or credit card details**.

🕒 **How to spot:** Check the sender's address; don't click suspicious links.

Malware & Ransomware

Malicious software that infects a website or a user's device. It can **steal data, lock files, or take down** an entire online store.

🛡️ **Protection:** Use antivirus, keep software updated.

Payment Fraud

Stolen credit cards or fake identities used to make purchases. The real owner disputes the charge, and the store **loses both money and goods**.

🛒 **Prevention:** Use CVV, 3D Secure, and address verification.

DDoS Attacks

Distributed Denial of Service 🕒 attackers flood a website with so much traffic that it **crashes**. Real customers can't access the store.

🛡️ **Mitigation:** Use firewalls, rate limiting, and CDN services.

Data Breaches

Hackers break into the store's database and **steal customer records** 🕒 names, emails, addresses, and even passwords.

🛡️ **Prevention:** Encrypt data, restrict access, and monitor logs.

E-Skimming

Cybercriminals inject malicious code into a checkout page to **capture payment details** in real time without the store knowing.

</> **Defense:** Regularly scan your website for unauthorized scripts.